SIMATSENGINEERING
ASSESSMENT TOOL 4

COURSENAME:ArtificialIntelligence **COURSECODE:CSA17**

COURSEOUTCOMECOVERED

CO3:*Analyzeknowledgerepresentationandreasoningusingpropositionallogic,FOL,inference,resolution, chainingmethods,knowledgerepresentationtoperformlogicalreasoningfordrawingvalidconclusionsand decision making. (BL4)*

AssessmentToolWeightage: 20%

TimeDuration:1Hour
QUESTIONS: 5 **TotalMarks:30**

AnnexureA

ResolutionAlgorithmImplementation

CodeofConduct:

I M.VAISHNAV(192524251) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
Iunderstandthatanyviolationofacademicintegrityruleswillresultindisciplinaryaction.

Signatureofthestudent: **M.VAISHNAV(192524251*)***

AnnexureA

ResolutionAlgorithmImplementation

Question1–RainandWet Ground

KnowledgeBase

1. Ifitrains,thenthegroundbecomes wet.
2. Itis raining.

Goal

Provethat**thegroundiswet**usingtheResolutionAlgorithm.

Tasks

1. Representthestatementsinpropositionallogic.
2. ConvertallstatementsintoCNF.

Negatethegoal.

1. ApplytheResolutionAlgorithmstepbystep.
2. ShowtheEmptyClause(□)ifthegoalisproved.

Question 2 – Student Assignment Submission

Knowledge Base

1. Ifastudentsubmitstheassignment,thenthestudentreceivesinternalmarks
2. Rahulsubmittedtheassignment.

Goal

Provethat**Rahulreceivesinternalmarks**usingtheResolutionAlgorithm. Tasks

1. Writethepropositionallogicexpressions.
2. ConvertthepremisesintoCNF.

Negatethegoal.

1. ApplytheResolutionAlgorithm.
2. Concludewhetherthegoalisproved.

Question 3 – Library Membership

Knowledge Base

1. Ifapersonisalibrarymember,thenthepersoncanborrowbooks.
2. Priyaisalibrarymember.

Goal

1. Provethat**Priyacanborrowbooks**usingtheResolutionAlgorithm.

Tasks

1. Representtheknowledgebaseusingpropositionallogic.
2. ConverteachstatementintoCNF.

Negate the goal.

1. Performresolutionuntilnonewclausescanbegenerated.
2. State the final conclusion.

Question4–PlacementEligibility

Knowledge Base

1. Ifastudentclearstheaptitudetest,thenthestudentiseligibleforplacement.
2. Arunclearedtheaptitudetest.

Goal

1. Provethat**Aruniseligibleforplacement**usingtheResolutionAlgorithm.

Tasks

1. Convertthestatementsintopropositionallogic.
2. WritetheCNFclauses.
3. Negatethegoal.
4. ApplytheResolutionAlgorithmstepbystep.
5. Drawtheresolutiontreeandconclude.

Question5–AccessControlSystem

KnowledgeBase

1. Ifauserentersthecorrectpassword,thentheuserisauthenticated.

2. Ifauserisauthenticated,thentheuserisgrantedaccess.
3. Theuserenteredthecorrectpassword.

Goal

Provethat**theuserisgrantedaccess**usingtheResolutionAlgorithm.

Tasks

1. Representtheknowledgebaseinpropositionallogic.
2. ConvertallimplicationsintoCNF.
3. Negatethegoal.
4. ApplytheResolutionAlgorithmtoderivetheEmptyClause(□).
5. Explaineachresolutionstepandwritethefinalconclusion.

1.

RUBRICSFORCALCULATION

NumericalProblems					
Criteria	**Weightage**	**Level4(Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem,identifiesall parameters and requirements accurately	Understands problemwith minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formulawith strong justification	Selectscorrect method with minor errors	Inappropriateor partiallycorrect method	Unable to selectmethod
Solution Process	25%	Logical, systematic steps with correct approachthroughout	Mostlycorrect steps with minormistakes	Disorganized steps;multiple errors	Noclear process
Accuracyof Calculation	20%	All calculations accurate;correctfinal answer	Minor calculation errors	Frequenterrors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineeringrelevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO3 – KNOWLEDGE REPRESENTATION AND REASONING

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QUESTION 1 – RAIN AND WET GROUND
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Knowledge Base:
1. If it rains, then the ground becomes wet.
2. It is raining.

Goal:
Prove that the ground is wet using the Resolution Algorithm.

Step 1: Propositional Logic Representation

R = It is raining
W = Ground is wet

Statements:
1. $R \rightarrow W$
2. R

Goal:
W

Step 2: Convert to CNF

$R \rightarrow W$
$= \neg R \lor W$

Clauses:
C1 = $(\neg R \lor W)$
C2 = (R)

Step 3: Negate the Goal

Goal = W

Negated Goal:
C3 = $(\neg W)$

Step 4: Apply Resolution

Resolve C1 and C3

$(\neg R \lor W)$
$(\neg W)$

$(\neg R)$

New Clause:
C4 = $(\neg R)$

Resolve C4 and C2

(¬R)
(R)

□

Step 5: Conclusion

Since the Empty Clause (□) is obtained, the goal is proved.

Therefore, the ground is wet.

==
QUESTION 2 – STUDENT ASSIGNMENT SUBMISSION
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Knowledge Base:
1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal:
Prove that Rahul receives internal marks.

Step 1: Propositional Logic Representation

S = Rahul submitted the assignment
M = Rahul receives internal marks

Statements:
1. S → M
2. S

Goal:
M

Step 2: Convert to CNF

S → M
= ¬S ∨ M

Clauses:
C1 = (¬S ∨ M)
C2 = (S)

Step 3: Negate the Goal

Goal = M

Negated Goal:
C3 = (¬M)

Step 4: Resolution

Resolve C1 and C3

(¬S ∨ M)
(¬M)

(¬S)

New Clause:
C4 = (¬S)

Resolve C4 and C2

(¬S)
(S)

□

Step 5: Conclusion

The Empty Clause (□) is obtained.

Therefore, Rahul receives internal marks.

==
QUESTION 3 – LIBRARY MEMBERSHIP
==

Knowledge Base:
1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal:
Prove that Priya can borrow books.

Step 1: Propositional Logic Representation

L = Priya is a library member
B = Priya can borrow books

Statements:
1. L → B
2. L

Goal:
B

Step 2: Convert to CNF

L → B

= ¬L ∨ B

Clauses:
C1 = (¬L ∨ B)
C2 = (L)

Step 3: Negate Goal

Goal = B

Negated Goal:
C3 = (¬B)

Step 4: Resolution

Resolve C1 and C3

(¬L ∨ B)
(¬B)

(¬L)

New Clause:
C4 = (¬L)

Resolve C4 and C2

(¬L)
(L)

□

Step 5: Conclusion

Empty Clause (□) is obtained.

Therefore, Priya can borrow books.

==
QUESTION 4 – PLACEMENT ELIGIBILITY
==

Knowledge Base:
1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal:
Prove that Arun is eligible for placement.

Step 1: Propositional Logic Representation

A = Arun cleared aptitude test

P = Arun is eligible for placement

Statements:
1. $A \rightarrow P$
2. A

Goal:
P

Step 2: Convert into CNF

$A \rightarrow P$
$= \neg A \lor P$

Clauses:
C1 = $(\neg A \lor P)$
C2 = (A)

Step 3: Negate Goal

Goal = P

Negated Goal:
C3 = $(\neg P)$

Step 4: Resolution

Resolve C1 and C3

$(\neg A \lor P)$
$(\neg P)$

$(\neg A)$

New Clause:
C4 = $(\neg A)$

Resolve C4 and C2

$(\neg A)$
(A)

□

Step 5: Resolution Tree

```
(¬A ∨ P)
   |
 (¬P)
   |
 (¬A)
   |
```

(A)
|
□

Step 6: Conclusion

The Empty Clause (□) is obtained.

Therefore, Arun is eligible for placement.

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QUESTION 5 – ACCESS CONTROL SYSTEM
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Knowledge Base:
1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal:
Prove that the user is granted access.

Step 1: Propositional Logic Representation

P = User entered correct password
A = User authenticated
G = User granted access

Statements:
1. $P \rightarrow A$
2. $A \rightarrow G$
3. P

Goal:
G

Step 2: Convert into CNF

$P \rightarrow A$
$= \neg P \lor A$

$A \rightarrow G$
$= \neg A \lor G$

Clauses:

C1 = $(\neg P \lor A)$
C2 = $(\neg A \lor G)$
C3 = (P)

Step 3: Negate Goal

Goal = G

Negated Goal:
C4 = (¬G)

Step 4: Resolution

Resolve C2 and C4

(¬A ∨ G)
(¬G)

(¬A)

New Clause:
C5 = (¬A)

Resolve C1 and C5

(¬P ∨ A)
(¬A)

(¬P)

New Clause:
C6 = (¬P)

Resolve C6 and C3

(¬P)
(P)

□

Step 5: Explanation

1. Converted all implications into CNF.
2. Negated the goal.
3. Applied resolution repeatedly.
4. Derived the Empty Clause (□).

Step 6: Conclusion

Since the Empty Clause (□) is obtained, the goal is proved.

Therefore, the user is granted access.

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FINAL CONCLUSION
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Q1 Result: Ground is Wet ✓

Q2 Result: Rahul Receives Internal Marks ✓

Q3 Result: Priya Can Borrow Books ✓

Q4 Result: Arun is Eligible for Placement ✓

Q5 Result: User is Granted Access ✓

In all five problems, the Resolution Algorithm produced the Empty Clause (□), proving that the given goals are logically valid.